

LOCAL_RUNTIME_REMOVAL

Local Runtime Removal — All-Local-Inference (llama.cpp + Ollama + SSH-remote + distributed) Removal & Redirect-to-Bridge Map

Revision: 1 **Last modified:** 2026-06-15T00:00:00Z **Status:** DESIGN / RESEARCH (uncommitted) — READ-ONLY survey, no source edited, no commit. FACT-cited per §11.4.6. **Authority:** Operator decision 2026-06-15 — FORBID ALL local runtimes (llama.cpp AND Ollama AND SSH-remote-llama.cpp/distributed). Every component uses ONLY LLMsVerifier-obtained (API) strongest verified models via pkg/bridge.BestTranslator (bridge built in parallel stream; see docs/design/LLMSVERIFIER_BRIDGE.md). **Scope:** This is the §11.4.145 8-angle impact-research + §11.4.150 deep-research **removal+redirect map** for the implementation stream. It does NOT implement. The bridge doc covers llama.cpp at site granularity; this doc EXTENDS the scope to Ollama + SSH + distributed and supplies the 8-angle GO/NO-GO, the gate plan, the Obsolete list, and the phased checklist.

0. TL;DR for the conductor

- **Local-runtime non-test source sites:** ~46 across three runtimes — **llama.cpp** ≈ 25 (19 enumerated in the bridge doc §2 + the 6 cmd/translate-ssh / cmd/ssh-translation llama-only clusters), **Ollama** ≈ 13, **SSH/distributed-coordination** ≈ 8 (provider-routing arms + worker config). Test cluster ≈ 20 files.
- **8-angle verdict: GO with conditions.** The redirect-to-bridge path is sound; the removal is bounded. Two NO-GO-until-resolved risks need operator input (R1 distributed/SSH = whole end-user capability removal under §11.4.122; R2 offline/no-key CI determinism — bridge needs keys, removing local fallback removes the only no-key path).
- **Components needing redirect to bridge.BestTranslator: 9** — unified-translator, cmd/server(+pkg/api/handler.go createTranslator), grpc-server, preparation-translator, markdown-translator, ebook-translator, cmd/cli, cmd/translator, pkg/api/batch_handlers.go. Plus pkg/coordination/multi_llm.go (Ollama discovery arm) and pkg/distributed/coordinator.go (local-provider priority arm).

- **Obsolete (§11.4.90) — local-ONLY binaries with no API translate path (FACT):** `cmd/translate-ssh`, `cmd/ssh-translation`. Both translate exclusively via remote `llama.cpp`; no `executeAPI/NewLLMTranslator/VerifiedFactory` path exists in either (`grep` returned empty). Reason superseded-by-design-change. Operator pre-approved removal/obsolete via the bridge-doc D2; this doc confirms the FACT that they are local-only.
 - **Operator-decision points (surface, do NOT decide):** **R1** — distributed-mode + SSH workers are a distinct end-user capability (§11.4.122): remove entirely vs redirect workers to call the verified API? **R2** — offline/no-key CI: removing local fallback means CI with NO `*_API_KEY` has NO translate path; confirm the honest path is SKIP-with-reason (§11.4.3), never a re-introduced silent local fallback. Both already align with bridge-doc D1/D2/D-ollama — this doc upgrades **D-ollama to a firm REMOVE** per the explicit operator decision.
-

1. Per-runtime site inventory (FACT, file:line cited)

1.1 llama.cpp (≈25 non-test source sites)

Authoritatively enumerated in `docs/design/LLMSVERIFIER_BRIDGE.md` §2 (19 sites: `llamacpp.go`, `llamacpp_provider.go`, `llm.go`:35/71/246/269–270, `unified-translator flags+executeLlamaCppTranslation`, `cmd/cli`, `cmd/markdown-translator`, `cmd/translator`, `cmd/ebook-translator`, `pkg/api/handler.go`:573/1814, `pkg/api/batch_handlers.go`, `pkg/grpc/*`, `pkg/grpc/proto/translator.pb.go`, `pkg/distributed/coordinator.go+pairing.go`, `pkg/challenge_runner/provider_challenges.go`, `pkg/report/report_generator.go`). **This doc adds the SSH llama-only cluster** (bridge-doc #8/#9 expanded):

- `cmd/translate-ssh/main.go` — `LlamaConfig`
`llm.LlamaCppProviderConfig` (:42), `default BinaryPath:"/usr/local/bin/llama.cpp"` (:201–202), `step3TranslateMarkdown shells/install remote llama.cpp` (:295–296, 647–810 — uploads `setup_llama.py/test_llama.py/install_llamacpp.sh`, probes `/usr/local/bin/llama.cpp` etc.). **No API translate path.**
- `cmd/ssh-translation/main.go` — workers hardwired `Type:"llamacpp"`, `BinaryPath:"/usr/local/bin/llama.cpp"` (:133–144). **No API translate path.**
- `cmd/translator/main.go`:36 — `LlamaConfig`
`llm.LlamaCppProviderConfig`.
- Type defs (`llamacpp_provider.go`:15–16, 47, 58, 94):
`LlamaCppProviderConfig`, `LlamaCppWorker`, `NewLlamaCppProvider`.

1.2 Ollama (≈13 non-test source sites, FACT)

#	Site (file:line)	Role	Action
O1	<code>pkg/translator/llm/ollama.go</code> (whole file)	<code>OllamaClient</code> , <code>localhost:11434 /api/generate</code>	DELETE (§11.4.124 git-history first)
O2		<code>ProviderOllama const</code>	REMOVE

#	Site (file:line)	Role	Action
	pkg/translator/llm/ llm.go:34		
O3	pkg/translator/llm/ llm.go:70	ValidModels [ProviderOllama] row	REMOVE
O4	pkg/translator/llm/ llm.go:245–246	"Ollama/LlamaCpp custom- model warn"	REMOVE (whole arm; both gone)
O5	pkg/translator/llm/ llm.go:267–268	case ProviderOllama: factory branch	REMOVE ; unknown → error
O6	cmd/unified-translator/ main.go:46,618,702–703,743	provider doc + case "mock", "ollama" no-key arm	REMOVE ollama from doc/help; case "ollama" → error/redirect
O7	cmd/cli/ main.go:669,679,721–722	ollama help + --disable-local- llms flag + example	REMOVE ollama refs; --disable- local-llms becomes no-op/ removed
O8	pkg/api/handler.go:567–571	"name":"ollama" in models list	REMOVE entry
O9	pkg/api/handler.go:1814	validProviders includes ollama	REMOVE from array
O10	pkg/api/ batch_handlers.go:122,234	case ... "ollama" ... valid- provider switch	REMOVE ollama (and llamacpp) arm
O11	pkg/coordination/ multi_llm.go:278–283	Ollama discovery (OLLAMA_ENABLED/ OLLAMA_MODEL)	REMOVE the Ollama provider block + disableLocalLLMs gate
O12	pkg/challenge_runner/ provider_challenges.go:32	ollama/llamacpp in challenge provider set	REMOVE both
O13	pkg/distributed/ coordinator.go:231	case "ollama", "llamacpp": priority arm	REMOVE/redirect (see R1)

Honest-boundary FACT (§11.4.6, NOT a removal site): pkg/storage/
redis.go:303–304 is a **comment** documenting the Ollama :-in-model-id cache-
key collision fix; the fix code stays (it is provider-agnostic length-prefixing). Do
NOT remove the fix; the comment may be left or reworded.

1.3 SSH-remote + distributed (≈8 coordination sites, FACT)

- cmd/unified-translator/main.go:304–305,692–694,627 — case "ssh"
translate branch + --ssh-* flags + key-check arm. **Redirect or remove per R1.**
- pkg/distributed/coordinator.go:206–245 — provider discovery (parses
worker providers incl. ollama/llamacpp) + getPriorityForProvider local
arm (:231). **Redirect/remove per R1.**
- pkg/distributed/pairing.go:37 — LocalLLMs []string ("ollama,
llamacpp models") worker field. **REMOVE** field per R1.

- pkg/grpc/proto/translator.pb.go:199 + pkg/grpc/{core_translator.go,server.go} — provider enum comment lists ssh/ollama/llamacpp. **REGENERATE proto** after enum edit (do NOT hand-edit .pb.go).
- Worker configs internal/working/config.worker.llamacpp.json + config.worker.ollama.json — both "default_provider":"llamacpp" with model_path/n_gpu_layers/gguf (FACT). **DELETE** (local-only fixtures). config.distributed*.json reviewed per R1.
- tests/websocket_monitoring_test.go:768 — Type:"ssh-llamacpp" worker fixture. **REWRITE/REMOVE** per R1.

version_manager is GENERIC (FACT): pkg/distributed/version_manager.go grep for llama|ollama|gguf|local-model returned EMPTY — it is binary-version-sync/rollback machinery, NOT llama-bound. It survives if distributed mode is redirected (R1 = redirect); it becomes Obsolete only if distributed mode is removed wholesale.

1.4 Test cluster (remove/rewrite under §11.4.43/§11.4.115)

llama: llamacpp_test.go, llamacpp_provider_test.go, llamacpp_buildargs_defect_test.go + arms in llm_test.go, provider_clients_test.go, integration_test.go, cmd/unified-translator/ssh_command_test.go, cmd/cli/main_test.go, cmd/markdown-translator/main_test.go. ollama: ollama_test.go, ollama_options_defect_test.go + arms in llm_test.go, provider_clients_test.go, cmd/cli/main_test.go, cmd/markdown-translator/main_test.go, cmd/translate-ssh/main_test.go, pkg/coordination/multi_llm_test.go+multi_llm_fault_test.go, pkg/distributed/coordinator_test.go+coordinator_translate_race_test.go+pairing_query_success_test.go, pkg/translator/universal_test.go, pkg/storage/storage_w2{0,1}_cachekey_*_test.go (cache-key tests — KEEP, rephrase ollama example only), test/unit/coordination_test.go (Ollama discovery tests).

2. Per-component redirect-to-bridge map

For each component currently able to select a local provider, the exact route is: replace the NewLLMTranslator(config) / executeLLamaCppTranslation / executeSSHTranslation provisioning with bridge.BestTranslator(ctx, task) (returns the strongest verified API model + fallback chain). Default provider becomes the bridge's strongest verified model, NOT a hardcoded provider string.

Component	Current local hook	Redirect
cmd/unified-translator	executeProviderTranslation switch ssh/llamacpp (:303-311); default -provider openai	Default path → bridge.BestTranslator; delete case "ssh"/case "llamacpp"; executeAPITranslation becomes the bridge-backed path

Component	Current local hook	Redirect
cmd/server + pkg/api/handler.go	createTranslator (handler.go:722-762); provider list/validProviders	createTranslator → bridge.BestTranslator; drop ollama/llamacpp from lists
grpc-server (pkg/grpc/*)	provider-string routing incl. ssh/local	route via bridge; regenerate proto enum
preparation-translator	llm.NewLLMTranslator (main.go:175), default deepseek	swap to bridge.BestTranslator (keeps API-only already; no local path today)
cmd/markdown-translator	llm.NewLLMTranslator (:314), default deepseek, case "llamacpp" (:286)	bridge.BestTranslator; remove llamacpp arm
cmd/ebook-translator	PROVIDER:llamacpp detection (:321)	bridge.BestTranslator; remove llamacpp detection arm
cmd/cli	-provider ... ollama/llamacpp, -disable-local-llms	bridge.BestTranslator; remove local flags/help/examples
cmd/translator	LlamaConfig (:36)	bridge.BestTranslator; remove LlamaConfig
pkg/api/batch_handlers.go	valid-provider switch (:122,234)	drop ollama/llamacpp; route via bridge
pkg/coordination/multi_llm.go	Ollama discovery arm (:278-283)	remove Ollama block; verified providers only
pkg/distributed/coordinator.go	local-provider priority/discovery	per R1 (remove or redirect to verified API)

pkg/models vs pkg/hardware (§11.4.124 dead-code FACT):

- pkg/models is imported by cmd/server/main.go:17, pkg/api/handler.go:11, pkg/security/user_auth.go:12 — **NOT dead** after local removal. KEEP.
- pkg/hardware is imported ONLY by pkg/translator/llm/llamacpp.go (non-test). After llama removal it is a dead-code candidate — but **investigate via git log -S/-G + check reflection/build-tag/codegen refs BEFORE deletion** (§11.4.124); separate removal-only commit citing git-history evidence. Do NOT delete on sight.

3. §11.4.145 8-angle impact GO/NO-GO

Angle	Finding (FACT/evidence)	Verdict
1. Correctness	Removing local providers removes valid factory branches; NewLLMTranslatorWithConfig case ProviderOllama/ProviderLlamaCpp	GO

Angle	Finding (FACT/evidence)	Verdict
	deleted → unknown-provider must error cleanly (existing default arm). Bridge returns a real verified client.	
2. Regression — defaults	config.json default_provider:"openai" (API) — unaffected. BUT worker configs config.worker. {llamacpp,ollama}.json default to llamacpp (FACT) — these break post-removal → DELETE them. markdown-translator default deepseek (API) OK. preparation API-only OK. Each component's default MUST become the bridge strongest model.	GO (with worker-config delete)
3. Latent/contract	Factory switch callers, validProviders arrays (handler.go:1814, batch_handlers.go:122/234), proto enum (translator.pb.go:199) all enumerate the local providers — every consumer must be edited in lockstep or a request naming ollama/llamacpp falls through to a now-error path. LlamaCppProviderConfig type referenced by cmd/translate-ssh + cmd/translator — removing the type breaks their compile (handled by their own removal/redirect).	GO (lockstep edit required)
4. Security	Net-positive: removes subprocess exec of llama-cli (llamacpp.go) + remote binary upload/install over SSH (translate-ssh uploads+runs install_llamacpp.sh) — reduces RCE/supply-chain surface. Bridge keeps keys in-env only (§11.4.10).	GO (improves)
5. Performance	Removes on-device GPU/CPU inference option; all inference becomes network API calls (latency + rate-limit dependent). Bridge has score-descending fallback chain (mitigation). No regression for the already-API default path.	GO
6. Host/data safety	Removes host subprocess spawning + remote-host binary install (§11.4.133-adjacent risk reduction). No data-loss path.	GO (improves)
7. Cross-feature	config.json/worker configs, pkg/distributed version_manager (generic, survives), cache-key fix in redis.go (KEEP — provider-agnostic). OLLAMA_ENABLED env contract removed (coordination + tests).	GO
8. Business-logic	Tests/Challenges DO assume local providers: test/unit/coordination_test.go Ollama-discovery tests, tests/websocket_monitoring_test.go ssh-	CONDITIONAL — R2

Angle	Finding (FACT/evidence)	Verdict
	llamacpp fixture, llamacpp/ollama unit tests. §11.4.50 determinism: bridge needs *_API_KEY → offline/no-key CI has NO translate path after local removal (R2).	

Overall: GO with two flagged risks (R1, R2) requiring operator input before Phase "remove".

Unmitigated risks needing operator input

- **R1 — distributed/SSH = whole end-user capability (§11.4.122).** cmd/translate-ssh, cmd/ssh-translation, distributed mode, and -provider ssh are a distinct shipped capability (remote-worker translation). Removing them is a §11.4.122 component removal → MUST be operator-confirmed. Choice: **(A)** remove entirely (Obsolete feature-removed) vs **(B)** keep distributed mode but redirect workers to call the verified API (version_manager + ssh_pool survive, only the llama-exec worker body is replaced). FACT: both SSH binaries are llama-ONLY today, so (A) deletes them; (B) requires building an API-worker body that does not exist yet.
- **R2 — offline/no-key CI determinism (§11.4.3 + §11.4.50 + §11.4.98).** Removing local fallback removes the only translate path that needs no API key. Deep research (§11.4.150, sources below) confirms the correct API-only pattern: tests that need a real model run only when keys are present and **SKIP-with-reason** otherwise (§11.4.3) — never a silent re-introduced local fallback (which would re-violate the operator decision). Confirm: the integration/e2e/Challenge suites adopt SKIP-when-no-key; unit tests use the existing pkg/translator/llm/mock.go (mocks permitted in unit tests only per §11.4.27).

4. §11.4.150 deep-research (≥2 cited latest-source angles)

Angle A — API-only architecture + graceful degradation (no local fallback):

Current (2025–2026) production practice for API-only LLM systems replaces *local* fallback with an **intelligent routing layer**: provider/model fallback chains + circuit breakers + token-bucket rate-limiting, returning a user-friendly degraded message rather than crashing. This is exactly the bridge's score-descending fallback chain (buildFallbackChain, bridge-doc §4.2) — i.e. removing local runtimes and relying on a verified-provider fallback chain is the documented best practice, NOT a missing-feature risk. ([sitepoint Hybrid Cloud-Local LLM Architecture Guide 2026](#), [getmaxim LLM Gateways 2026](#), [truefoundry rate-limiting AI agents](#))

Angle B — CI testing of LLM-dependent code without keys/offline (the R2 risk): The documented pattern is NOT a local model fallback but a **test mode that skips the external provider call** while exercising everything else, plus **record/replay** for deterministic real-behaviour without per-run API spend, plus **keys via CI secrets/env only**. Maps directly to: SKIP-with-reason when no key (§11.4.3), mock.go for unit contracts (§11.4.27), and real-provider integration tests gated on key presence. The honest handling of R2 is therefore SKIP-with-reason —

confirmed by external precedent, not a silent local fallback. ([dev.to testing LLM integrations in CI without burning tokens](#), [Red Hat integration testing fast-moving AI backend](#))

Conclusion: no bigger hidden problem — the design (verified-provider fallback chain + SKIP-when-no-key CI) matches current best practice. The only genuine trade-off is "no offline inference," which IS the operator's explicit decision.

5. Obsolete list (§11.4.90 — operator-gated removal)

Binary/asset	FACT
cmd/translate-ssh	llama-on grep emp
cmd/ssh-translation	llama-on hardware
internal/working/config.worker.llamacpp.json	default
internal/working/config.worker.ollama.json	default
internal/working/config.distributed*.json	distributed
pkg/translator/llm/llamacpp.go, llamacpp_provider.go, ollama.go	impl file
pkg/hardware	imported
internal/scripts/ {setup_llama.py, test_llama.py, install_llamacpp.sh, translate_llm_only.py}	uploaded
pkg/models, pkg/distributed/version_manager.go, ssh_pool.go, pkg/storage/redis.go cache-key fix: NOT Obsolete (live non-local consumers).	

6. Gate plan — CM-N0-LOCAL-RUNTIME (§1.1 paired-mutation)

Generalises the bridge-doc CM-N0-LLAMACPP to cover all three runtimes:

- **Pre-build gate CM-N0-LOCAL-RUNTIME:** grep the authoritative source tree (cmd/ + pkg/ + internal/, EXCLUDING submodules llms_verifier/ helix_qa/ vision_engine/, EXCLUDING removal-test files + the mock.go unit seam) for the forbidden set: llama.cpp llamacpp llama-cli

ProviderLlamaCpp ProviderOllama OllamaClient :11434
OLLAMA_ENABLED LlamaCppProviderConfig, and (if R1=A)
executeSSHTranslation/translate-ssh. Any NEW occurrence → FAIL.

- **Paired §1.1 mutation:** re-introduce one forbidden usage (e.g. a ProviderOllama const) → assert gate FAILs; restore → assert PASS.
 - **Per-component RED-first redirect tests (§11.4.115):** for each redirected component, a RED_MODE=1 test reproducing "component still provisions a local client" on the pre-fix artifact, flipped RED_MODE=0 to the standing guard asserting it provisions via bridge.BestTranslator (verified API). Registered into the §11.4.135 regression suite.
 - **No-key SKIP guard (§11.4.3):** a test asserting that with NO *_API_KEY present every translate entry point returns an honest SKIP-with-reason / "API key required" error — NEVER a local fallback (proves R2 handled honestly; paired mutation re-adds a local fallback → gate FAILs).
 - **Worker-config gate:** assert no tracked config carries "default_provider":"llamacpp" / "ollama" / gguf / model_path.
-

7. Phased ordered implementation checklist (the implementation stream executes this)

Pre-req: pkg/bridge exists with BestTranslator (parallel stream, bridge-doc Phases 0–2). Do not start Phase R-3 until bridge.Open+BestTranslator build green.

- **Phase R-0 — Operator gate (BEFORE any removal).** AskUserQuestion R1 (distributed/SSH remove-vs-redirect) + R2 confirm (SKIP-when-no-key, no silent fallback) + confirm D-ollama=REMOVE. (§11.4.122/§11.4.66/§11.4.101).
- **Phase R-1 — Redirect components (additive, locals still present, builds green).** Route the 9 components (+coordination) through bridge.BestTranslator; make it the default provisioning path. Each component independently buildable. §11.4.115 RED-first redirect test per component.
- **Phase R-2 — Remove Ollama (§11.4.124 git-history first; separate commit).** O1–O13: delete ollama.go, const/ValidModels/warn-arm/factory-branch, lists (handler/batch), coordination arm, challenge set, distributed arm; remove -disable-local-llms/OLLAMA_ENABLED contract; rewrite/remove ollama tests; reword redis.go comment.
- **Phase R-3 — Remove llama.cpp (§11.4.124 + §11.4.122).** Bridge-doc §2 sites #1–#7,#10–#17,#19 + LlamaCppProviderConfig type; regenerate proto; delete worker configs; investigate pkg/hardware orphaning (separate removal-only commit with git-history evidence).
- **Phase R-4 — SSH/distributed per R1.** (A) Obsolete cmd/translate-ssh+cmd/ssh-translation+distributed fixtures + the internal/scripts/*llama* set + -provider ssh + distributed local arms; OR (B) redirect workers to verified API (build API-worker body; keep version_manager/ssh_pool). Track each removal as §11.4.90 Obsolete.
- **Phase R-5 — Gate + full retest.** Add CM-N0-LOCAL-RUNTIME + paired mutation + no-key SKIP guard + worker-config gate. config.json already API-default (no change); update Documentation/AGENTS.md, README,

.golangci.yml if needed, the §11.4.131 session-resumption file. Full-suite retest (§11.4.40). docs/qa// transcript for the redirected translate path (§11.4.83).

8. Catalogue-check (§11.4.74) + cross-refs

- Catalogue-Check: reuse digital.vasic.translator/pkg/bridge (parallel stream) — every redirect calls BestTranslator, no reimplementation.
- Catalogue-Check: extend vasic-digital/LLMsVerifier — verified-model source unchanged.
- Composes with bridge-doc §2 (llama.cpp sites), §4 (bridge API), §7 (CM-N0-LLAMACPP → generalised here to CM-N0-LOCAL-RUNTIME), §9 (D1/D2/D-ollama → R1/R2 + firm D-ollama=REMOVE here).

Sources verified 2026-06-15

- <https://www.sitepoint.com/hybrid-cloudlocal-llm-the-complete-architecture-guide-2026/>
- <https://www.getmaxim.ai/articles/top-5-llm-gateways-in-2025-the-definitive-guide-for-production-ai-applications/>
- <https://www.truefoundry.com/blog/rate-limiting-ai-agents-preventing-llm-api-exhaustion>
- <https://dev.to/akarshc/how-to-test-llm-integrations-in-ci-without-burning-tokens-libh>
- <https://developers.redhat.com/articles/2026/05/27/how-we-built-integration-testing-fast-moving-ai-backend>